Chapter 4 Review

Campbell Biology, AP Edition, 12e

Chapter 4 Review

Summary of Key Concepts

Of all the elements in the periodic table, why is carbon central to all forms of life? **(BIG IDEA 2)**

Concept 4.1: Organic chemistry is key to the origin of life

- **Organic** compounds, once thought to arise only within living organisms, were finally synthesized in the laboratory.
- Living matter is made mostly of carbon, oxygen, hydrogen, and nitrogen. Biological diversity results from carbon's ability to form a huge number of molecules with particular shapes and properties.

How did Stanley Miller's experiments support the idea that, even at life's origins, physical and chemical laws govern the processes of life?

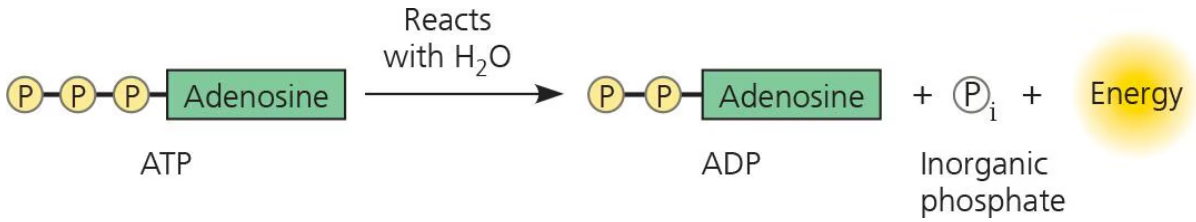
Concept 4.2: Carbon atoms can form diverse molecules by bonding to four other atoms

- Carbon, with a valence of 4, can bond to various other atoms, including O, H, and N. Carbon can also bond to other carbon atoms, forming the carbon skeletons of organic compounds. These skeletons vary in length and shape and have bonding sites for atoms of other elements.
- **Hydrocarbons** consist of carbon and hydrogen.
- **Isomers** are compounds that have the same molecular formula but different structures and therefore different properties. Three types of isomers are **structural isomers,*cis-trans*isomers**, and **enantiomers**.

VISUAL SKILLS*Refer back to*Figure 4.9*. What type of isomers are acetone and propanal? How many asymmetric carbons are present in acetic acid, glycine, and glycerol phosphate? Can these three molecules exist as forms that are enantiomers?**

Concept 4.3: A few chemical groups are key to molecular function

- Chemical groups attached to the carbon skeletons of organic molecules participate in chemical reactions (**functional groups)** or contribute to function by affecting molecular shape (see Figure 4.9).
- **ATP (adenosine triphosphate)** consists of adenosine attached to three phosphate groups. ATP can react with water or other molecules, forming ADP (adenosine diphosphate) and inorganic phosphate. This reaction releases energy that can be used by the cell.



The reactant A T P (an adenosine attached to 3 P's representing phosphate groups) reacts with water, producing P sub i (inorganic phosphate), A D P (adenosine attached to 2 P's) and energy.

*In what ways does a methyl group differ chemically from the other six important chemical groups shown in*Figure 4.9*?*